

NIKOLSK, Russian Federation

WMO: 270660

Lat: 59.534N

Lon: 45.461E

Elev: 143

StdP: 99.62

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.5	-28.1	-34.3	0.2	-31.4	-30.7	0.2	-28.0	6.4	-6.7	5.6	-6.6	0.9	10	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	28.9	19.8	27.0	18.8	25.1	17.8	21.1	26.8	20.0	25.3	19.0	23.4	2.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.2	23.6	18.1	13.2	22.3	17.1	12.4	21.5	61.5	26.9	58.0	25.7	54.4	23.4	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.9	5.2	4.5	-34.0	31.3	3.8	1.6	-36.8	32.5	-39.0	33.4	-41.2	34.3	-44.0	35.5		
			-34.1	22.3	3.8	1.3	-36.9	23.3	-39.1	24.0	-41.2	24.7	-44.0	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-11.9	-10.6	-4.4	3.2	10.4	15.0	17.8	15.2	9.7	3.1	-3.8	-8.6	(6)
(7)		DBStd	11.67	8.21	7.68	5.37	4.62	5.22	4.35	3.63	3.98	3.96	4.23	5.90	7.76	(7)
(8)		HDD10.0	3245	680	577	447	207	61	8	1	4	52	218	415	576	(8)
(9)		HDD18.3	5672	938	810	706	453	249	118	54	113	258	474	665	834	(9)
(10)		CDD10.0	689	0	0	0	4	74	157	243	165	44	3	0	0	(10)
(11)		CDD18.3	77	0	0	0	0	4	17	39	16	1	0	0	0	(11)
(12)		CDH23.3	799	0	0	0	1	69	194	378	153	4	0	0	0	(12)
(13)		CDH26.7	193	0	0	0	0	12	45	94	42	0	0	0	0	(13)
(14)	Wind	WSAvg	2.1	2.1	2.2	2.2	2.1	2.2	2.0	1.7	1.7	1.8	2.2	2.2	2.2	(14)
(15)	Precipitation	PrecAvg	606	37	27	29	36	51	75	81	67	54	58	48	45	(15)
(16)		PrecMax	818	69	54	63	84	131	177	190	140	121	103	117	86	(16)
(17)		PrecMin	458	7	0	4	10	17	14	24	14	10	0	12	13	(17)
(18)		PrecStd	86	15	13	13	19	27	45	38	31	24	22	20	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.5	2.3	10.0	22.0	28.4	31.1	31.3	30.8	23.7	16.6	8.6	4.5	(19)
(20)			MCWB	0.9	1.4	4.6	12.5	17.3	20.3	21.3	21.3	16.7	11.3	7.4	4.1	(20)
(21)		2%	DB	0.7	1.2	6.1	17.4	25.2	27.9	29.2	27.1	20.6	13.0	5.5	2.4	(21)
(22)			MCWB	0.3	0.5	2.2	9.8	15.5	18.9	20.4	19.5	15.2	10.6	4.6	1.9	(22)
(23)		5%	DB	0.0	0.5	4.0	14.2	22.4	25.6	27.4	24.5	18.1	10.9	4.0	1.1	(23)
(24)			MCWB	-0.4	-0.1	1.1	7.8	13.7	17.7	19.5	18.2	14.0	8.9	3.4	0.5	(24)
(25)		10%	DB	-1.4	-0.7	2.4	11.4	19.8	23.5	25.4	22.1	16.2	9.1	2.4	0.2	(25)
(26)			MCWB	-2.0	-1.4	0.5	6.4	12.4	16.8	18.7	17.1	13.0	7.7	1.8	-0.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.1	1.7	5.4	13.0	18.7	21.5	22.7	22.2	17.7	12.7	7.7	4.2	(27)
(28)			MCDWB	1.3	2.0	9.2	20.3	25.5	28.2	28.5	28.7	21.9	15.0	8.5	4.5	(28)
(29)		2%	WB	0.4	0.7	3.1	10.5	16.9	20.2	21.4	20.4	16.0	10.7	4.7	2.1	(29)
(30)			MCDWB	0.7	1.1	5.0	16.0	22.7	26.3	27.0	25.7	19.1	12.7	5.2	2.4	(30)
(31)		5%	WB	-0.5	0.0	1.8	8.6	15.1	18.9	20.4	18.9	14.6	9.2	3.3	0.7	(31)
(32)			MCDWB	0.0	0.5	3.3	13.2	20.8	23.7	25.6	22.7	17.6	10.7	3.9	1.0	(32)
(33)		10%	WB	-2.0	-1.4	0.8	6.7	13.5	17.6	19.5	17.7	13.3	7.8	1.9	-0.2	(33)
(34)			MCDWB	-1.5	-0.7	2.1	10.8	18.3	22.1	24.2	21.4	15.8	9.1	2.3	0.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.3	7.4	8.9	9.9	11.2	10.5	10.4	9.3	8.0	5.0	3.9	5.0	(35)
(36)			MCDBR	4.8	4.6	10.7	15.4	15.4	14.4	13.9	13.1	11.9	8.7	4.6	3.6	(36)
(37)			MCWBR	4.8	4.4	7.4	8.9	7.1	6.3	6.1	6.0	7.0	5.9	4.2	3.7	(37)
(38)		5% WB	MCDBR	4.6	4.3	7.5	13.8	13.3	12.1	11.9	11.2	10.1	7.8	4.3	3.8	(38)
(39)			MCWBR	4.8	4.2	5.6	8.6	6.8	5.9	5.9	5.8	6.2	5.8	4.2	3.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.265	0.305	0.339	0.380	0.361	0.359	0.393	0.385	0.350	0.329	0.289	0.246	(40)	
(41)		taud	2.043	2.060	2.055	2.153	2.421	2.449	2.366	2.365	2.446	2.363	2.148	1.996	(41)	
(42)		Ebn at Noon	497	673	772	812	868	874	830	807	779	666	470	369	(42)	
(43)		Edh at Noon	49	83	114	125	103	102	108	101	79	63	44	34	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.35	1.03	2.31	3.70	5.05	5.30	5.39	3.95	2.24	0.94	0.31	0.19	(44)	
(45)		RadStd	0.06	0.27	0.35	0.44	0.43	0.46	0.43	0.36	0.29	0.15	0.06	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	-199	N/A	N/A

Nomenclature: See separate page